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F. Reachability

time limit per test: 1.5 seconds

memory limit per test: 256 megabytes

You are given a **unweighted** graph with N cities and M **bidirectional** roads. The nodes are numbered from 1 to N . The graph contains no self-loops or multiple edges.

Your task is to answer Q queries of the form (x, y) , where you have to say if we can reach city y from city x using the roads.

Input

The first line contains three integers N , M and Q ($1 \leq N \leq 10^5, 1 \leq M \leq 10^5, 1 \leq Q \leq 10^5$) — the number of cities, total number of roads and total number of queries.

This is followed by M lines describing the M roads. The i -th line contains two integers u and v ($1 \leq u, v \leq N$), meaning that the i -th road connects city u and city v .

The final Q lines describe the queries. The i -th line contains two integers x and y ($1 \leq x, y \leq N$) describing the i -th query.

Output

For each query (x, y) , output YES if it is possible to reach y from x and output NO otherwise.

Examples

input	Copy
<pre>5 3 4 4 3 1 2 4 5 1 2 2 4 3 1 3 5</pre>	
output	Copy
<pre>YES NO NO YES</pre>	

input	Copy
<pre>8 7 4 6 2 7 5 7 8 3 1 2 3 2 1 3 4 3 1 4 6 7 2 4 6</pre>	
output	Copy

CSE221 Assignment 05 Summer 2026

Contest is running

4 weeks

Contestant



→ Submit?

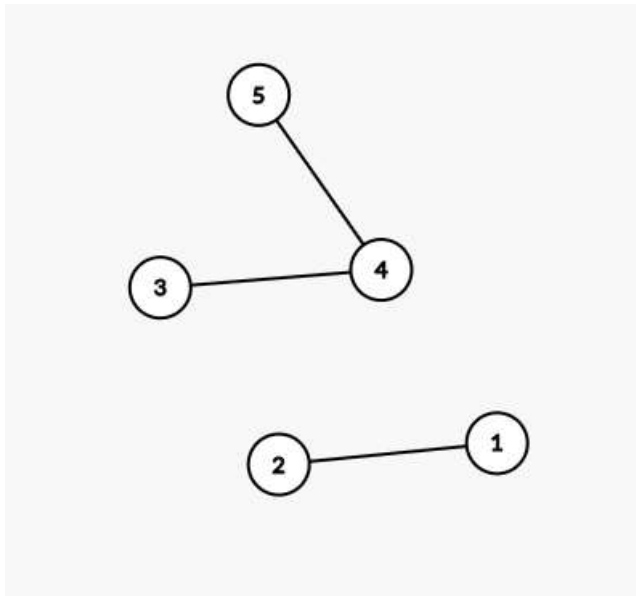
Language: Java 21 64bit Choose file: Choose File No file chosen

Success!



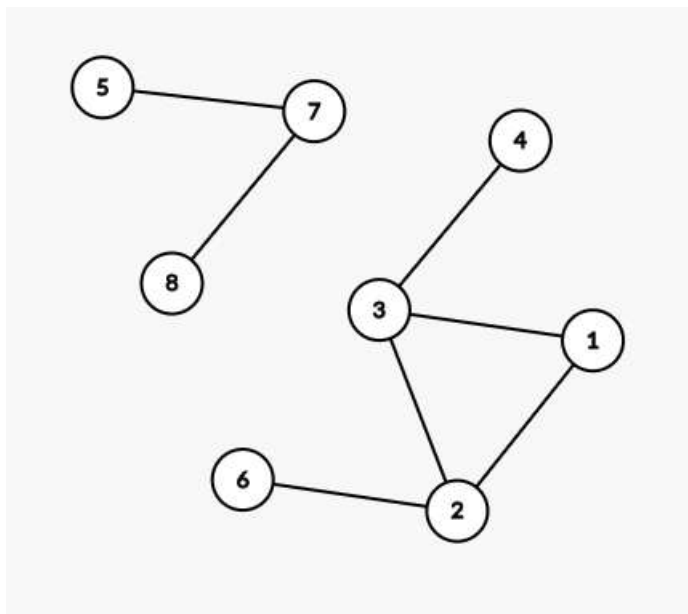
Submit

YES
YES
NO
YES

Note

The graph in the first sample.

The image above represents the graph for the first sample. The answer to the first query 1 2 is YES because we can obviously go to 2 from 1. The answer to the second query 2 4 is NO because we can not go from 2 to 4 in this graph.



The graph in the second sample.

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